

Computational Capital Budgeting: Simulation Design for Allocating Scarce AI Tokens

1 Outline: What Are We Testing?

1.1 Core Motivation

Firms increasingly encourage employees and departments to use generative AI tools. If AI use is priced by token consumption, then tokens become a scarce organizational resource. This creates a capital allocation problem: firms must decide how to allocate limited AI resources across departments, projects, and time.

The central idea is that AI tokens can be treated as a form of *computational capital*: a scarce input that firms consume to generate analysis, automation, decisions, forecasts, code, contracts, marketing content, and other productive outputs. Unlike cash, however, tokens are consumed when used, their productivity is uncertain, and their value depends heavily on task type, model quality, prompt quality, and organizational context.

1.2 Main Research Question

When AI tokens become a scarce productive resource inside firms, should they be allocated through fixed budgets, centralized capital budgeting, real-options reserve policies, auctions, or internal token markets?

1.3 Simple Version of the Question

If a firm has a limited number of AI tokens, how should it decide which departments and projects get to use them?

1.4 Theoretical Framing

The project connects to several existing finance theories:

1.5 Core Hypotheses

- H1: Market-based allocation mechanisms, such as internal token markets and auctions, generate higher firm value than equal or historical token budgets.
- H2: Centralized allocation performs well when headquarters has accurate information, but deteriorates when departments possess private information about project quality.
- H3: Real-options reserve policies become more valuable when future project arrivals, token prices, or model productivity are more uncertain.

Finance Theory	Connection to AI Token Allocation
Capital budgeting	Tokens are allocated to projects based on expected value creation.
Capital rationing	Firms may have more positive-value AI projects than available tokens.
Real options	Saving tokens preserves flexibility for future high-value opportunities or future model improvements.
Internal capital markets	Headquarters allocates scarce resources across divisions with different information and incentives.
Agency theory	Departments may exaggerate project value or token needs to obtain larger token allocations.
Mechanism design	Auctions or internal token markets may induce departments to reveal their true demand.
Production theory	Tokens are modeled as a factor of production that improves project quality with diminishing marginal returns.

H4: Token markets improve efficiency but may increase concentration, hoarding, or inequality across departments.

H5: Multi-agent AI workflows create interdependencies that make simple project-level NPV rules incomplete.

1.6 AI Automation vs. Manual Intervention

- **Can be automated using AI:** literature mapping, initial theory synthesis, generation of hypotheses, simulation coding, robustness checks, figure/table generation, drafting, and workflow documentation.
- **Needs manual intervention:** final choice of research question, assessment of whether the idea is sufficiently novel for the conference, validation of theoretical framing, interpretation of results, and decision about whether to pursue this project over alternatives.

2 Simulation Setup: Entities and Behavioral Assumptions

2.1 Simulated Firm

The simulation creates a synthetic firm that operates over repeated periods, such as weeks or months. The firm has a fixed total AI-token budget in each period. Departments receive stochastic project opportunities and compete for token allocations.

2.2 Departments

The baseline firm includes six departments:

1. Finance
2. Legal
3. Marketing

4. Research and Development
5. Operations
6. Human Resources

Each department differs in project arrival rate, project risk, expected payoff, token productivity, and information advantage.

Department	Project Type	Expected Behavior	Interpretation
Finance	Forecasting, valuation, reporting	High discipline, measurable returns	Projects are easier to evaluate using NPV-like rules.
Legal	Contract review, compliance, risk prevention	High urgency, downside protection	Value may come from avoiding losses rather than creating visible upside.
Marketing	Content, campaigns, customer analytics	High variance, high experimentation	Many projects, uncertain payoff, potential overuse of tokens.
R&D	Product research, coding, prototyping	High upside, long horizon	Some projects are lottery-like and resemble growth options.
Operations	Process automation, logistics	Medium risk, steady productivity	Token use may generate stable efficiency gains.
HR	Recruiting, training, internal support	Lower direct payoff, organizational value	Benefits may be diffuse and harder to measure.

2.3 Projects

Each department receives projects over time. A project is characterized by:

- potential payoff,
- token requirement,
- deadline or urgency,
- project risk,
- strategic importance,
- department-specific productivity,
- observability to headquarters,
- possibility of spillovers to other departments.

2.4 Token Productivity Function

Tokens are assumed to improve project quality with diminishing returns. For project i in department j at time t :

$$Q_{ijt} = 1 - \exp\left(-\alpha_j \frac{T_{ijt}}{N_{ijt}}\right), \quad (1)$$

where Q_{ijt} is project quality, α_j is department-specific AI productivity, T_{ijt} is tokens allocated, and N_{ijt} is the number of tokens required for full-quality completion.

Realized project value is:

$$\Pi_{ijt} = V_{ijt}Q_{ijt} - P_t T_{ijt}, \quad (2)$$

where V_{ijt} is potential project value and P_t is the token price.

2.5 How Entities Respond

- **Departments** rank projects by expected benefit per token, urgency, and strategic value. In agency settings, departments may exaggerate project value or token needs.
- **Headquarters/CFO** attempts to maximize firm value subject to the token budget constraint. The CFO observes noisy project estimates rather than true project values.
- **Internal market participants** buy, sell, or bid for tokens based on perceived marginal value.
- **AI agents** can be assigned to departments and instructed to make allocation decisions under different rules. Their choices can be compared against algorithmic benchmark policies.

2.6 AI Automation vs. Manual Intervention

- **Can be automated using AI:** generation of department profiles, parameter calibration scenarios, agent decision rules, simulation code, and descriptions of entity behavior.
- **Needs manual intervention:** choice of economically defensible parameter ranges, checking whether agent behavior is theoretically meaningful, and ensuring that department assumptions are not arbitrary.

3 Simulation Design

3.1 Time Structure

The baseline simulation runs over T periods, such as 52 weeks. Each simulated firm-year consists of repeated project arrivals, token allocation decisions, project execution, and payoff realization.

3.2 Project Arrivals

Projects arrive randomly by department:

$$K_{jt} \sim \text{Poisson}(\lambda_j), \quad (3)$$

where K_{jt} is the number of projects arriving in department j at time t and λ_j is the department-specific arrival rate.

3.3 Potential Project Value

Project values are drawn from a lognormal distribution:

$$V_{ijt} \sim \text{Lognormal}(\mu_{V,j}, \sigma_{V,j}). \quad (4)$$

This captures the idea that most projects have modest value, while a small number have very large upside.

3.4 Token Requirements

Token requirements are also drawn from a lognormal distribution:

$$N_{ijt} \sim \text{Lognormal}(\mu_{N,j}, \sigma_{N,j}). \quad (5)$$

This reflects the fact that most AI tasks are small, while complex multi-agent workflows may consume very large token budgets.

3.5 Payoff Risk

Realized project value includes idiosyncratic uncertainty:

$$\tilde{V}_{ijt} = V_{ijt} \exp(\varepsilon_{ijt}), \quad \varepsilon_{ijt} \sim \mathcal{N}(0, \sigma_{\varepsilon,j}^2). \quad (6)$$

3.6 Information Friction

Departments observe project quality more accurately than headquarters. Headquarters observes:

$$\hat{V}_{ijt} = V_{ijt} + \nu_{ijt}, \quad \nu_{ijt} \sim \mathcal{N}(0, \sigma_{\nu,j}^2). \quad (7)$$

Strategic reporting can be modeled as:

$$\hat{N}_{ijt} = N_{ijt}(1 + b_j), \quad (8)$$

where b_j is a department-level exaggeration or padding parameter.

3.7 Allocation Mechanisms

Each simulation run compares multiple allocation systems under the same project realizations.

1. **Equal Budgeting:** each department receives the same token budget.
2. **Historical Budgeting:** departments receive tokens based on prior-period usage.
3. **Centralized NPV Allocation:** headquarters ranks projects by expected value per token and funds the highest-ranked projects.
4. **Real-Options Reserve Policy:** headquarters withholds a fraction of tokens for future high-value project arrivals.
5. **Internal Token Market:** departments trade tokens using internal prices based on marginal value.

6. **Auction Allocation:** departments bid for scarce tokens, and tokens are allocated to the highest bidders.
7. **Hybrid Policy:** baseline allocation plus market-based reallocation after departments observe project demand.

3.8 Performance Metrics

The primary outcome is firm value per token:

$$\text{Value per Token} = \frac{\sum_{i,j,t} \Pi_{ijt}}{\sum_{i,j,t} T_{ijt}}. \quad (9)$$

Additional metrics include:

- total firm value,
- risk-adjusted firm value,
- unused tokens,
- missed high-value projects,
- project completion rate,
- value volatility,
- concentration of tokens across departments,
- token hoarding,
- welfare loss from private information,
- value of the real-options reserve.

3.9 Empirical Interpretation

Although the data are simulated, the design creates empirical evidence in the form of controlled computational experiments. The simulation allows us to vary uncertainty, information asymmetry, token scarcity, and market design while holding other features constant.

3.10 AI Automation vs. Manual Intervention

- **Can be automated using AI:** writing the simulation code, generating Monte Carlo runs, creating parameter grids, producing tables and plots, identifying anomalous results, and proposing robustness checks.
- **Needs manual intervention:** deciding which allocation mechanisms are economically meaningful, reviewing whether the simulation is too mechanical, and validating that results are not artifacts of parameter choices.

4 Expected Outcomes

4.1 Predicted Result 1: Fixed Budgets Are Inefficient

Equal or historical budgets are expected to perform poorly when project arrivals are uneven across departments. Some departments will leave tokens unused while others miss high-value opportunities.

4.2 Predicted Result 2: Centralized Budgeting Works Under Low Information Friction

When headquarters observes project value accurately, centralized NPV allocation should perform well. However, performance should decline as information noise and departmental strategic reporting increase.

4.3 Predicted Result 3: Internal Markets Improve Allocation but Create Side Effects

Internal token markets should improve allocation efficiency by moving tokens toward departments with higher marginal value. However, these markets may create token concentration, hoarding, or strategic behavior.

4.4 Predicted Result 4: Real-Options Reserves Matter Under Uncertainty

When future project arrivals are highly uncertain or when model productivity may improve, reserving tokens can increase firm value. This result would connect AI-token allocation to real-options theory.

4.5 Predicted Result 5: Token Scarcity Changes Organizational Behavior

When token budgets are loose, allocation mechanisms matter less. When tokens are scarce, the choice of allocation mechanism becomes more important.

4.6 Predicted Result 6: AI Workflows May Create Spillovers

In multi-agent settings, allocating tokens to one department or agent may increase the productivity of others. For example, an R&D agent may produce code or analysis that Legal, Operations, or Finance can reuse. This makes token budgeting different from standard project-level NPV ranking.

4.7 AI Automation vs. Manual Intervention

- **Can be automated using AI:** summarizing results, detecting patterns, suggesting explanations, generating alternative hypotheses, and drafting the results section.
- **Needs manual intervention:** assessing whether the outcomes are interesting, whether they are obvious, whether they would satisfy finance reviewers, and whether the project is worth pursuing relative to stronger alternatives.

5 Shocks

5.1 Project Demand Shock

A demand shock increases the number of project arrivals in one or more departments:

$$\lambda_{j,t} = \lambda_j(1 + s_{j,t}). \quad (10)$$

Interpretation: sudden litigation, product launch, regulatory filing, crisis response, or marketing campaign.

5.2 Firm-Wide Productivity Shock

A common productivity shock affects all departments:

$$A_t \sim \mathcal{N}(0, \sigma_A^2). \quad (11)$$

Interpretation: macroeconomic shock, organizational disruption, AI outage, or broad improvement in AI adoption.

5.3 Model Improvement Shock

At random times, model productivity improves:

$$M_t = \begin{cases} 1 + \Delta_M, & \text{with probability } p_M, \\ 1, & \text{otherwise.} \end{cases} \quad (12)$$

This shock increases α_j , making each token more productive. This is central to the real-options logic: if future models may become better, there may be value in preserving token flexibility.

5.4 Token Price Shock

Token prices follow a geometric random walk with possible jump shocks:

$$P_t = P_{t-1} \exp(\eta_t + J_t), \quad \eta_t \sim \mathcal{N}(0, \sigma_P^2). \quad (13)$$

where J_t is a jump term capturing sudden price increases due to compute scarcity or provider price changes.

5.5 Information Shock

The precision of headquarters' information varies over time:

$$\sigma_{\nu,t} = \sigma_{\nu}(1 + z_t). \quad (14)$$

Interpretation: some periods are easier to evaluate than others. For example, Finance projects may be measurable, while Legal or R&D projects may be difficult to value.

5.6 Strategic Behavior Shock

Departments may become more aggressive in overstating token demand when budgets are tight:

$$b_{j,t} = b_j + \phi \cdot \text{Scarcity}_t. \quad (15)$$

This creates an agency problem: departments have incentives to inflate demand when tokens are scarce.

5.7 Spillover Shock

Some projects create reusable outputs that improve productivity elsewhere:

$$\alpha_{k,t+1} = \alpha_{k,t} + \rho_{jk} \cdot \mathbb{1}\{\text{project } i \text{ completed in department } j\}. \quad (16)$$

Interpretation: one AI project may create templates, code, workflows, or knowledge that increases other departments' productivity.

5.8 AI Automation vs. Manual Intervention

- **Can be automated using AI:** coding shock processes, running sensitivity analyses, identifying which shocks drive the results, and creating robustness heatmaps.
- **Needs manual intervention:** deciding which shocks are central rather than decorative, making sure shock assumptions are not arbitrary, and choosing which results to emphasize.

6 Expected Deliverables

6.1 Research Deliverables

1. A conceptual framework defining AI tokens as computational capital.
2. A theoretical model of computational capital budgeting under uncertainty.
3. An agent-based simulation comparing allocation mechanisms.
4. Monte Carlo evidence on which mechanisms maximize firm value under different conditions.
5. Robustness tests varying scarcity, uncertainty, information asymmetry, model improvement, and token price volatility.
6. Documentation of what parts of the research workflow were automated by AI and where human judgment was required.

6.2 Tables and Figures

Potential paper outputs include:

- table of simulation parameters,
- comparison of allocation mechanisms,
- value per token across mechanisms,

- heatmap of mechanism performance under scarcity and uncertainty,
- token concentration across departments,
- value of real-options reserves,
- welfare loss from information asymmetry,
- flow diagram of the AI-automated research workflow.

6.3 AI Workflow Deliverables

Because the target conference values AI-generated research, the project should also produce a reproducible workflow appendix:

1. initial human prompt,
2. AI agent roles,
3. model configurations,
4. data and simulation generation process,
5. code generation history,
6. validation and debugging log,
7. robustness-check log,
8. human intervention log,
9. final paper generation process.

6.4 AI Automation vs. Manual Intervention

- **Can be automated using AI:** generation of the simulation appendix, code repository structure, LaTeX tables, figures, draft paper, workflow documentation, and replication package.
- **Needs manual intervention:** deciding whether the paper has enough novelty, approving final theoretical claims, validating finance relevance, checking code correctness, and deciding whether this project should be the main submission.